

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

<think> To solve this problem, let's first calculate how much of the file download was completed before the Windows update forced a restart. Since the total file size is 200 GB and 40% of the way through the download corresponds to $(200 \times 0.40 = 80)$ GB, we know that 80 GB of the file were downloaded before the interruption. Next, we need to account for the 20 minutes the download was paused due to the update. During these 20 minutes, no data was downloaded. Now, let's calculate how much of the file was left to download after the interruption. Initially, the remaining part of the file to be downloaded was $(200 - 80 = 120)$ GB. Then, we need to consider that after the restart, Carla has to start downloading the file from the beginning again. Therefore, we add the 80 GB already downloaded back to the 120 GB that need to be downloaded from the start. So, the total amount of data that needs to be downloaded now is $(80 + 120 = 200)$ GB. </think> <answer> 200 </answer>

